

Auteur: Abdellah El Moussaoui
Studierichting: Informatica
Oefeningenreeks 1C nr. 45.10

Trek de derdemachtswortels in $\mathbb{C}$:

$$z^3 = 4\sqrt{2}(1 + i)$$

Voor $1 + i$ hebben we:

$$r = \sqrt{1^2 + 1^2} = \sqrt{2}$$

$$\theta = \text{Bgtan} \left(\frac{1}{1} \right) = \frac{\pi}{4}$$

$$\Rightarrow 1 + i = \sqrt{2} \text{cis} \left(\frac{\pi}{4} \right)$$

$$\Rightarrow z^3 = 4\sqrt{2} \cdot \sqrt{2} \text{cis} \left(\frac{\pi}{4} \right) = 8 \text{cis} \left(\frac{\pi}{4} \right)$$

$$\Rightarrow z_k = 2 \text{cis} \left(\frac{\frac{\pi}{4} + 2k\pi}{3} \right), \quad k = 0, 1, 2$$

$$\Rightarrow z_k = 2 \text{cis} \left(\frac{\pi}{12} + \frac{2k\pi}{3} \right), \quad k = 0, 1, 2$$

$$\Rightarrow z_0 = 2 \text{cis} \left(\frac{\pi}{12} \right) = \frac{\sqrt{6} + \sqrt{2}}{2} + i \frac{\sqrt{6} - \sqrt{2}}{2}$$

$$\Rightarrow z_1 = 2 \text{cis} \left(\frac{3\pi}{4} \right) = -\sqrt{2} + i\sqrt{2}$$

$$\Rightarrow z_2 = 2 \text{cis} \left(\frac{17\pi}{12} \right) = -\frac{\sqrt{6} - \sqrt{2}}{2} - i \frac{\sqrt{6} + \sqrt{2}}{2}$$

$$\Rightarrow \boxed{z_0 = \frac{\sqrt{6} + \sqrt{2}}{2} + i \frac{\sqrt{6} - \sqrt{2}}{2}, \quad z_1 = -\sqrt{2} + i\sqrt{2}, \quad z_2 = -\frac{\sqrt{6} - \sqrt{2}}{2} - i \frac{\sqrt{6} + \sqrt{2}}{2}}$$

References